EXAMPLE NOTE

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Abstract. This example demonstrates the basic structure of the note template: sections, theorem-like environments, cross-references, displayed equations, lists, commutative diagrams, citations, and a bibliography.

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1. A first section

This is ordinary mathematical text. We cite a standard reference [Har77] and refer to Theorem 1.2 below. Later we also use the lowercase form proposition 1.3.

Definition 1.1. A *sample object* is an object introduced only to demonstrate the formatting of definitions.

Theorem 1.2. *Let X be a sample object. Then the following properties hold:*

(i) X has a first property.
(ii) X has a second property.

Proof. This is a sample proof. The displayed equation

$$(a+b)^2 = a^2 + 2ab + b^2$$

illustrates ordinary mathematical display style. □

By Item (i), the first property holds.

Proposition 1.3. *The sample object of Definition 1.1 also satisfies a proposition.*

Lemma 1.4. *This is a sample lemma.*

Corollary 1.5. *This is a sample corollary.*

Remark 1.6. This is a remark. It demonstrates the remark style.

Example 1.7. This is an example environment.

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2. A diagram

The template currently supports commutative diagrams using `tikz-cd`:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D. \end{array}$$

3. References

We have used Definition 1.1, theorem 1.2, and proposition 1.3 and cited [Har77].

References

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977.